

TASK 2: BASIC WIRESHARK TRAFFIC CAPTURE AND ANALYSIS

Submitted by

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1.OBJECTIVE

The objective of this task is to capture and analyze network traffic using Wireshark. The captured packets were examined to identify common networking protocols, understand how devices communicate over a network, and study the flow of data between a client and server. This exercise helped in understanding real-world network communication and basic traffic analysis techniques used in cybersecurity.

2.INTRODUCTION TO WIRESHARK

Wireshark is a free and open-source network protocol analyzer used to monitor and inspect network traffic in real time. It captures packets travelling across a network and allows users to analyze communication between devices. Wireshark is widely used by network administrators, security analysts, and cybersecurity professionals for troubleshooting, protocol analysis, and security investigations.

Key Features of Wireshark

- Captures live network traffic
- Supports thousands of protocols
- Displays packet details in a readable format
- Provides filtering and search capabilities
- Helps identify network issues and security threats

3. SYSTEM REQUIREMENTS

Hardware Requirements

- Computer/Laptop
- Internet Connection

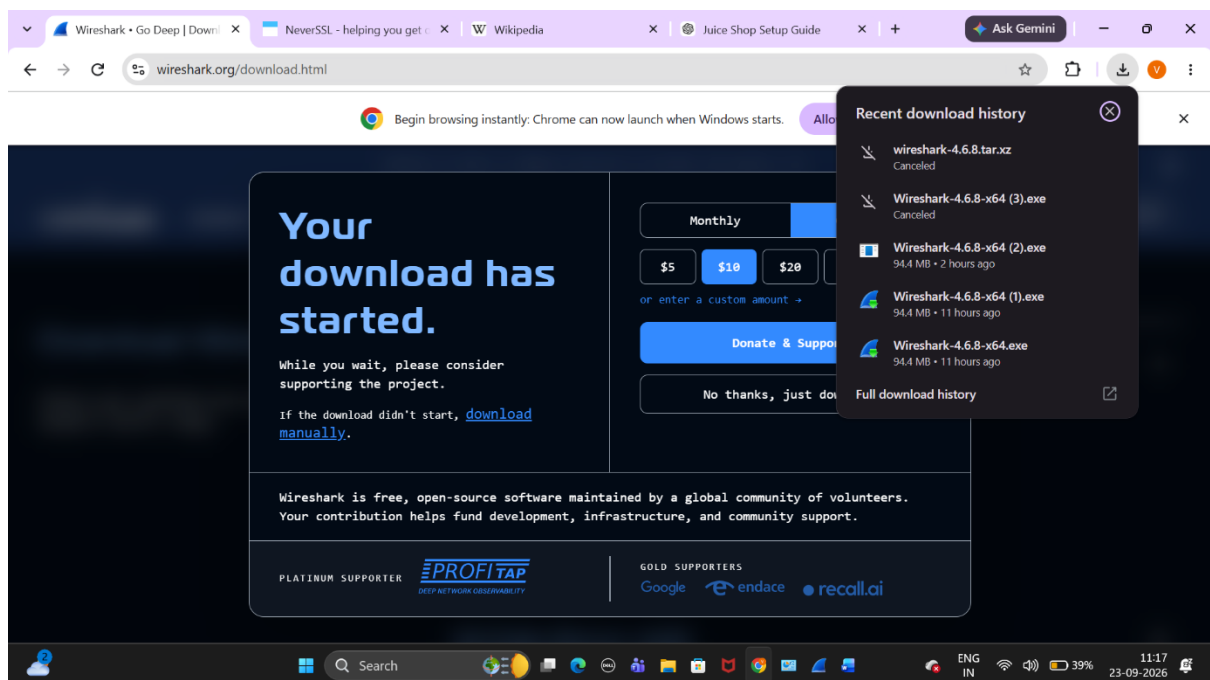
Software Requirements

- Windows 10/11
 - Wireshark
 - Web Browser
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4. INSTALLATION OF WIRESHARK

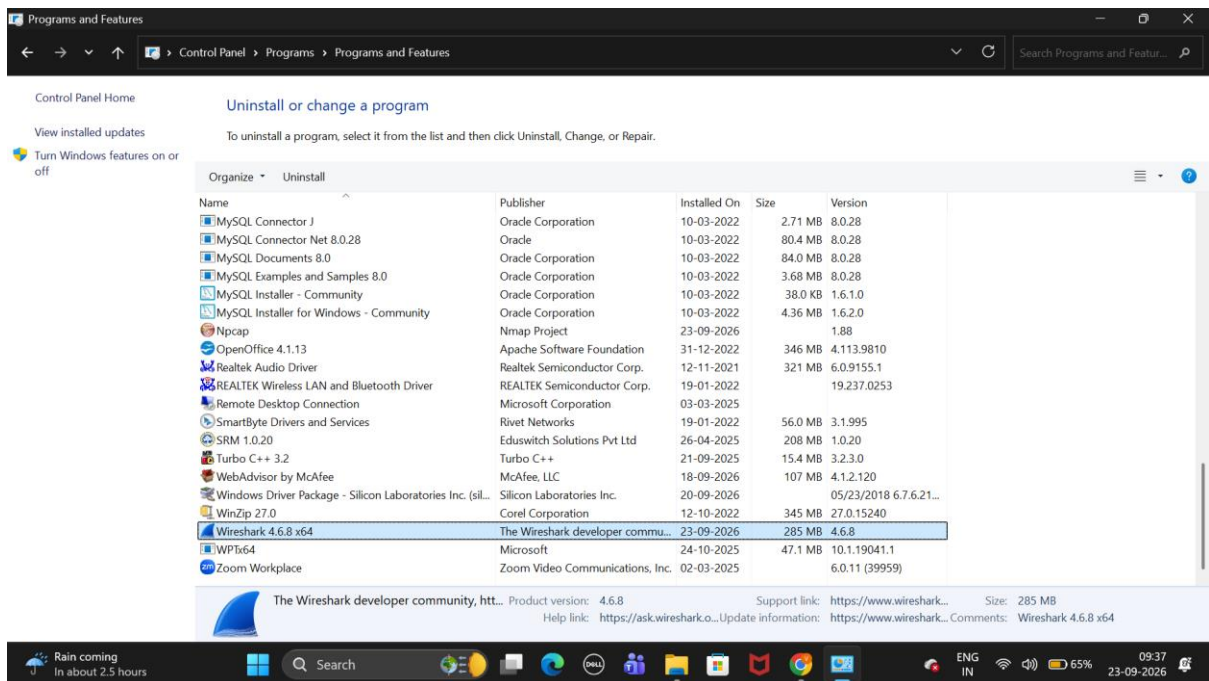
Step 1

Downloaded Wireshark from the official website.



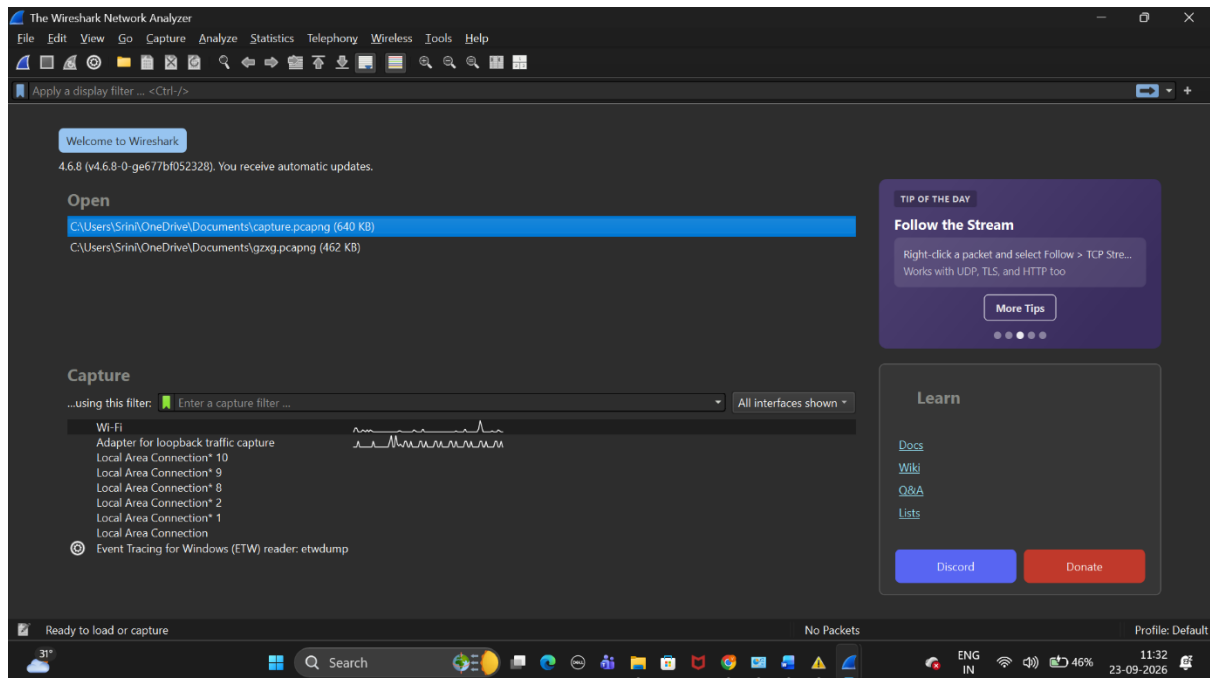
Step 2

Installed Wireshark and selected the default installation options.



Step 3

Launched Wireshark successfully and selected the active network interface



5. TRAFFIC CAPTURE PROCEDURE

Step 1: Starting Packet Capture

After launching Wireshark, the active Wi-Fi network interface was selected from the list of available interfaces. Packet capture was started by double-clicking the selected interface. Once the capture began, Wireshark started recording all incoming and outgoing network packets in real time.

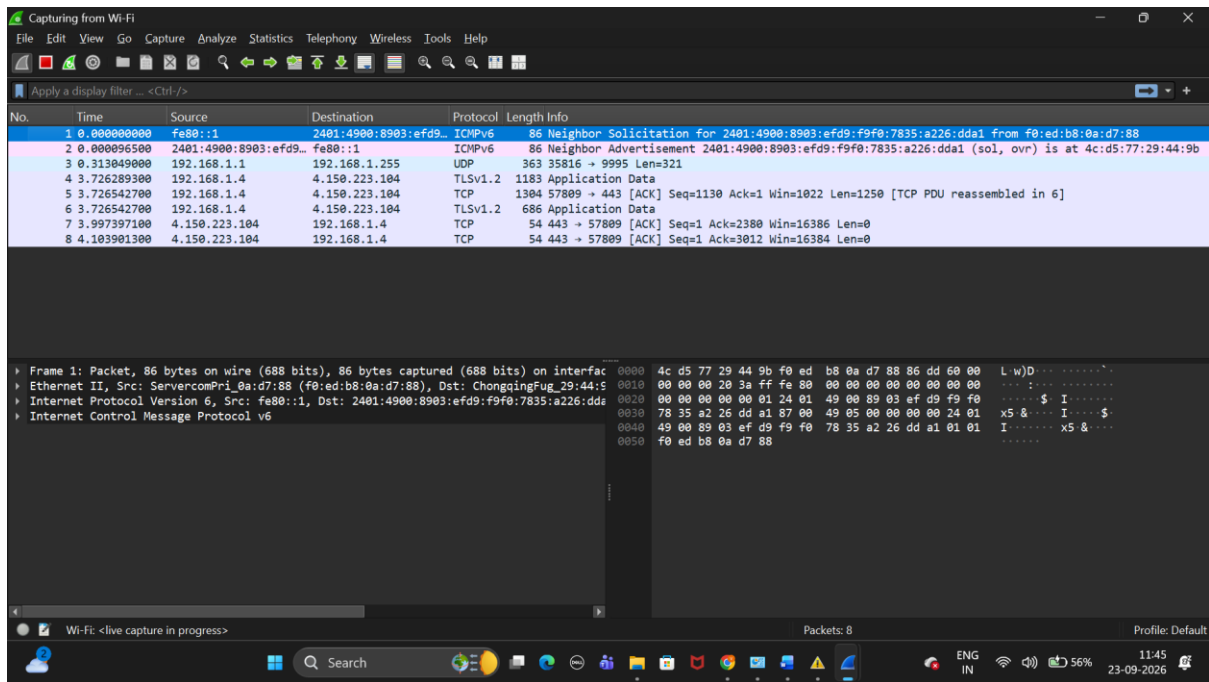


Figure 4: Starting packet capture on the active Wi-Fi interface.

Step 2: Generating Network Traffic

To generate network traffic for analysis, several websites were opened in a web browser. Activities such as browsing webpages, refreshing tabs, and performing online searches created network packets that could be captured by Wireshark. This provided a variety of protocols for analysis.

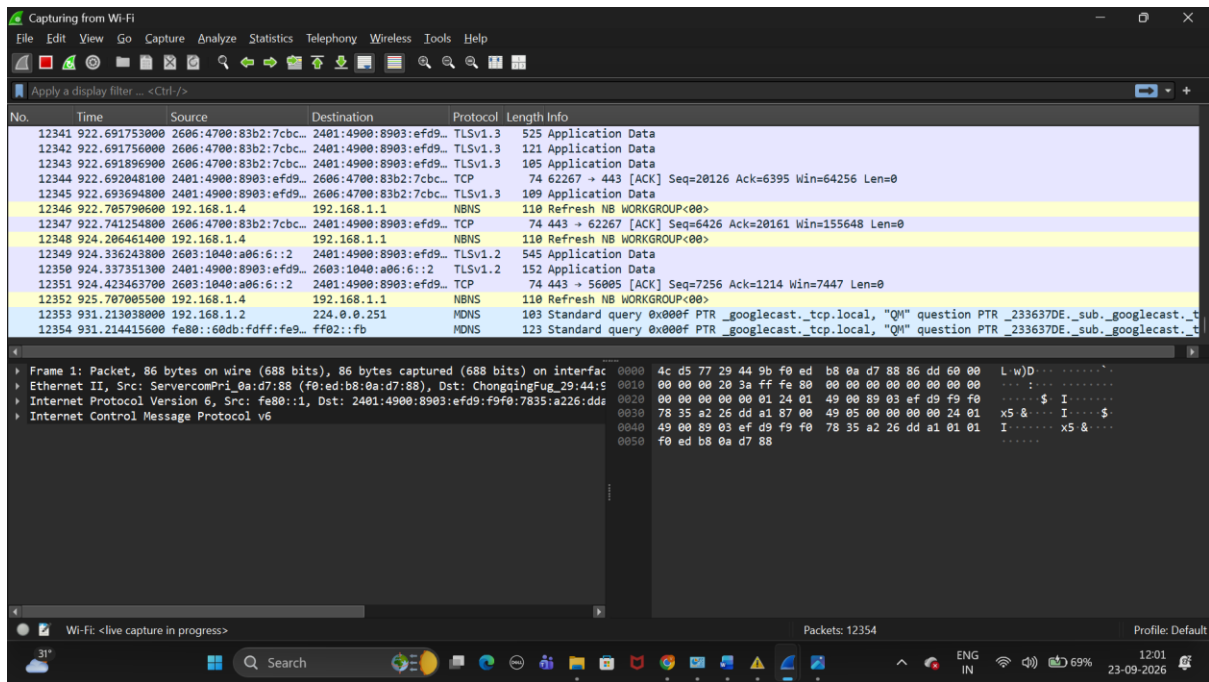


Figure 5: Live network traffic captured by Wireshark.

Step 3: Stopping Packet Capture

After sufficient traffic was collected, the packet capture process was stopped using the Stop Capture button. The captured data was then saved as a packet capture file (.pcapng) for further analysis.

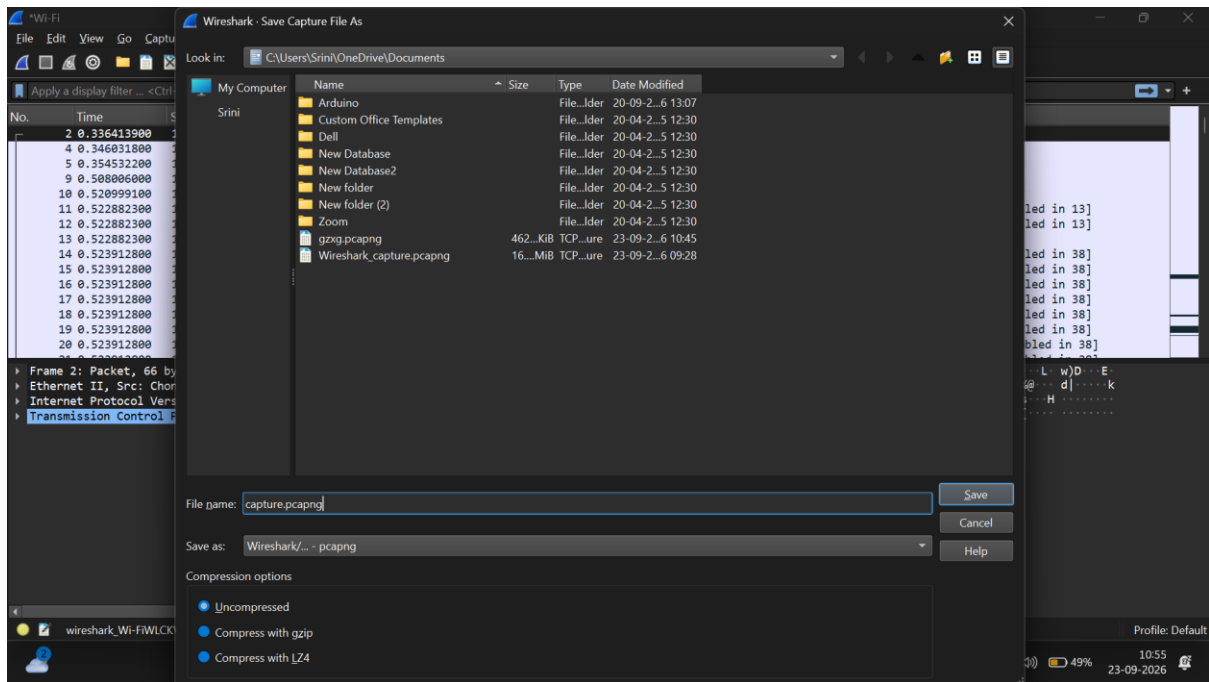


Figure 6: Captured traffic saved for analysis.

6. PROTOCOL IDENTIFICATION AND ANALYSIS

Overview

During the traffic capture process, several network protocols were observed. These protocols play different roles in network communication. The captured packets were analyzed to understand how devices communicate and exchange information across the network.

6.1 DNS (Domain Name System)

Purpose

DNS is responsible for converting domain names such as www.google.com into IP addresses that computers can understand.

Observation

DNS packets were observed whenever a website was accessed. Before connecting to a website, the system first sent a DNS request to obtain the corresponding IP address.

Importance

DNS makes internet usage easier because users can access websites using names instead of numerical IP addresses.

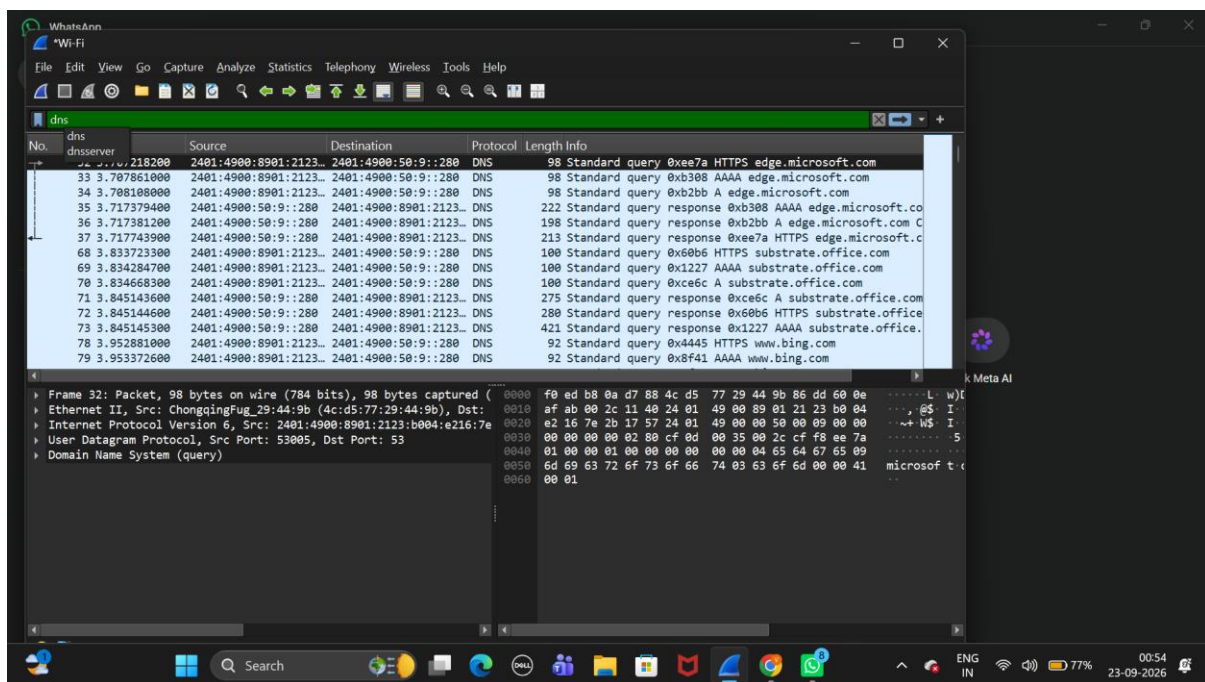


Figure 7: DNS request and response packets.

6.2 TCP (Transmission Control Protocol)

Purpose

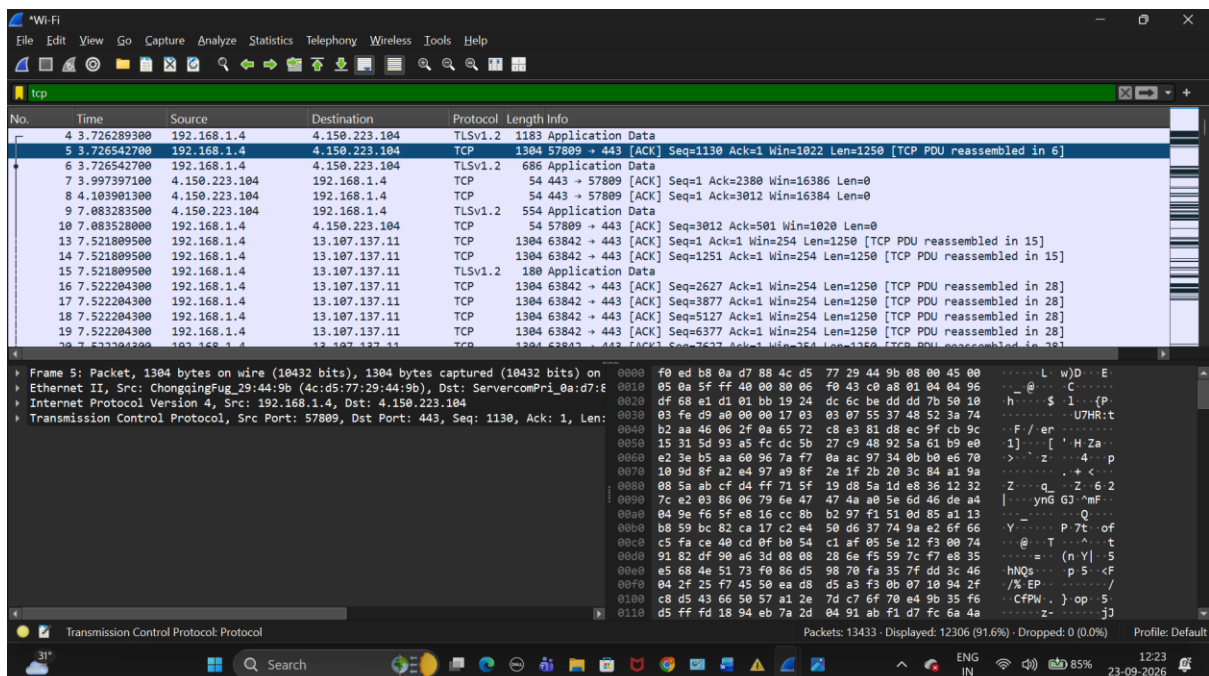
TCP provides reliable communication between devices. It ensures that packets are delivered correctly and in the proper sequence.

Observation

Multiple TCP packets were captured during web browsing activities. TCP connections were established before data transfer occurred.

Importance

TCP is widely used because it guarantees reliable and error-checked data transmission.



The image shows a Wireshark packet capture window titled "Wi-Fi". The packet list pane on the left shows a list of captured packets. The packet details pane on the right shows the details of the selected packet (No. 5). The packet bytes pane at the bottom shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
4	3.726289300	192.168.1.4	4.150.223.104	TLSv1.2	1183	Application Data
5	3.726542700	192.168.1.4	4.150.223.104	TCP	1304	57809 → 443 [ACK] Seq=1130 Ack=1 Win=1022 Len=1250 [TCP PDU reassembled in 6]
6	3.726542700	192.168.1.4	4.150.223.104	TLSv1.2	686	Application Data
7	3.997397100	4.150.223.104	192.168.1.4	TCP	54	443 → 57809 [ACK] Seq=1 Ack=2380 Win=16384 Len=0
8	4.103901300	4.150.223.104	192.168.1.4	TCP	54	443 → 57809 [ACK] Seq=1 Ack=3012 Win=16384 Len=0
9	7.083283500	4.150.223.104	192.168.1.4	TLSv1.2	554	Application Data
10	7.083528000	192.168.1.4	4.150.223.104	TCP	54	57809 → 443 [ACK] Seq=3012 Ack=501 Win=1020 Len=0
13	7.521809500	192.168.1.4	13.107.137.11	TCP	1304	63842 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1250 [TCP PDU reassembled in 15]
14	7.521809500	192.168.1.4	13.107.137.11	TCP	1304	63842 → 443 [ACK] Seq=1251 Ack=1 Win=254 Len=1250 [TCP PDU reassembled in 15]
15	7.521809500	192.168.1.4	13.107.137.11	TLSv1.2	180	Application Data
16	7.522204300	192.168.1.4	13.107.137.11	TCP	1304	63842 → 443 [ACK] Seq=2627 Ack=1 Win=254 Len=1250 [TCP PDU reassembled in 28]
17	7.522204300	192.168.1.4	13.107.137.11	TCP	1304	63842 → 443 [ACK] Seq=3877 Ack=1 Win=254 Len=1250 [TCP PDU reassembled in 28]
18	7.522204300	192.168.1.4	13.107.137.11	TCP	1304	63842 → 443 [ACK] Seq=5127 Ack=1 Win=254 Len=1250 [TCP PDU reassembled in 28]
19	7.522204300	192.168.1.4	13.107.137.11	TCP	1304	63842 → 443 [ACK] Seq=6377 Ack=1 Win=254 Len=1250 [TCP PDU reassembled in 28]

Frame 5: Packet, 1304 bytes on wire (10432 bits), 1304 bytes captured (10432 bits) on 0
Ethernet II, Src: ChongqingFug_29:44:9b (4c:d5:77:29:44:9b), Dst: ServercomPri_0a:d7:8
Internet Protocol Version 4, Src: 192.168.1.4, Dst: 4.150.223.104
Transmission Control Protocol, Src Port: 57809, Dst Port: 443, Seq: 1130, Ack: 1, Len: 1250

0000 f0 ed b8 0a d7 88 4c d5 77 29 44 9b 08 00 45 00Lw)D...E
0010 05 0a 5f ff 40 00 00 06 f0 43 c0 a8 01 04 04 96 ..._@...C...
0020 df 68 e1 d1 01 bb 19 24 dc 6c be dd dd 7b 50 10 ...h...\$...l...{P
0030 03 fe d9 a0 00 00 17 03 03 07 55 37 48 52 3a 74 ...>...z...4...p
0040 b2 aa 46 06 2f 0a 65 72 c8 e3 81 d8 ec 9f cb 9c ...F...en...t
0050 15 31 5d 93 a5 fc dc 5b 27 c9 48 92 5a 61 b9 e0 ...I...[...H...Za...
0060 e2 3e b5 aa 60 96 7a f7 0a ac 97 34 0b 00 e6 70 ...>...z...4...p
0070 10 9d 8f a2 e4 97 a9 8f 2e 1f 2b 20 3c 84 a1 9a ...>...z...4...p
0080 08 5a ab cf d4 ff 71 5f 19 d8 5a 1d a8 36 12 32 ...Z...q...z...6.2
0090 7c e2 03 86 06 79 6e 47 47 4a a0 5e 6d 46 de a4 ...|...ynG...GJ...mf...
00a0 04 9e f6 5f e8 16 cc 8b b2 97 f1 51 0d 85 a1 13 ...>...z...4...p
00b0 b8 59 bc 82 ca 17 c2 e4 50 d6 37 74 9a e2 6f 66 ...Y...T...P...7...of
00c0 c5 fa ce 40 cd 0f b0 54 c1 af 05 5e 12 f3 00 74 ...@...T...t...
00d0 91 82 df 90 a6 3d 08 08 28 6e f5 59 7c f7 e8 35 ...>...z...4...p
00e0 a5 68 4e 51 73 f0 85 d5 90 70 fa 35 7f dd 3c 46 ...hno...p...5...f
00f0 04 2f 25 f7 45 50 aa d8 d5 a3 f3 0b 07 10 94 2f .../%...EP.../...
0100 c8 d5 43 66 50 57 a1 2e 7d c7 6f 70 e4 9b 35 f6 ...CFPW...op...5
0110 d5 ff df 18 94 eb 7a 2d 04 91 ab f1 d7 fc 6a 4a ...>...z...4...p

Figure 8: TCP communication observed during packet capture.

6.3 HTTPS / TLS

Write:

Purpose

HTTPS uses TLS encryption to secure communication between a web browser and a web server.

Observation

Encrypted HTTPS traffic was observed while accessing websites. The packet contents could not be directly read because the data was encrypted.

Importance

HTTPS protects user privacy by preventing attackers from viewing sensitive information transmitted across the network.

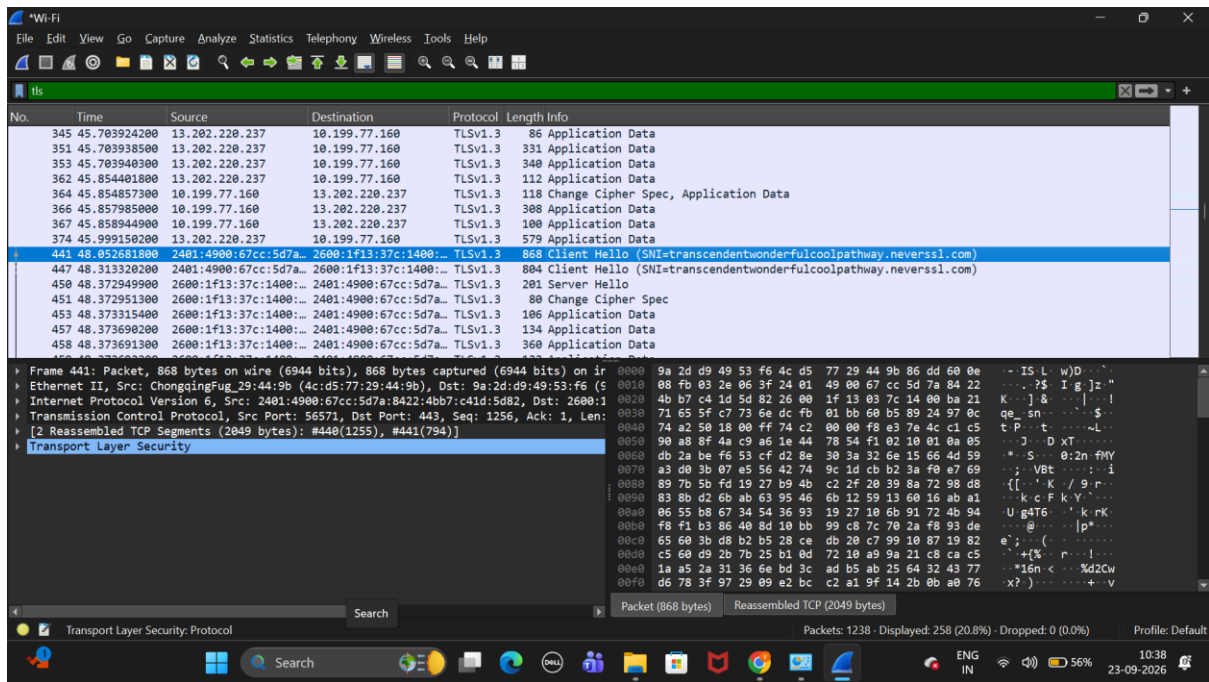


Figure 9: Encrypted HTTPS/TLS communication.

6.4 ARP (Address Resolution Protocol)

Purpose

ARP is used to map IP addresses to MAC addresses within a local network.

Observation

ARP requests and responses were observed during network communication.

Importance

ARP enables devices on a local network to locate and communicate with each other.

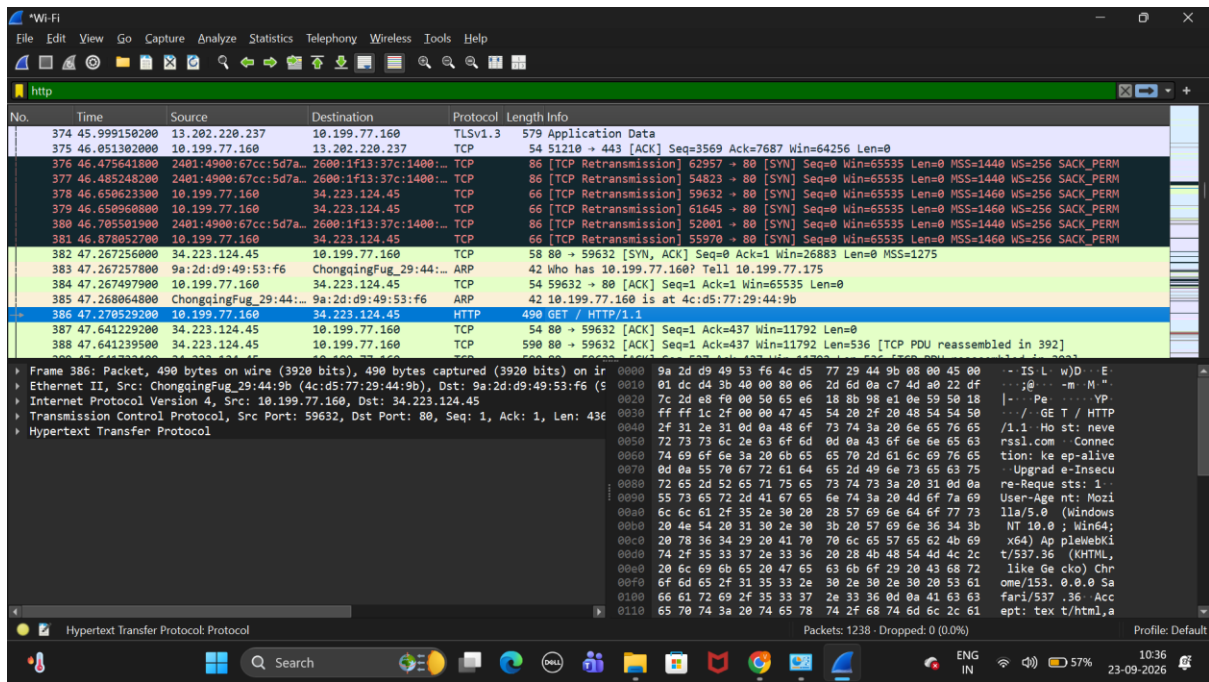


Figure 10: ARP request and response packets.

6.5 Packet-by-Packet Walkthrough (TCP 3-Way Handshake)

Packet 1: SYN

The client sends a SYN packet to initiate a connection with the server.

Packet 2: SYN-ACK

The server responds with a SYN-ACK packet, acknowledging the client's request and initiating its own synchronization.

Packet 3: ACK

The client sends an ACK packet to acknowledge the server's response, completing the connection establishment process.

Observation

The TCP three-way handshake was successfully observed before data transmission began.

Importance

The handshake ensures reliable communication and confirms that both devices are ready to exchange data.

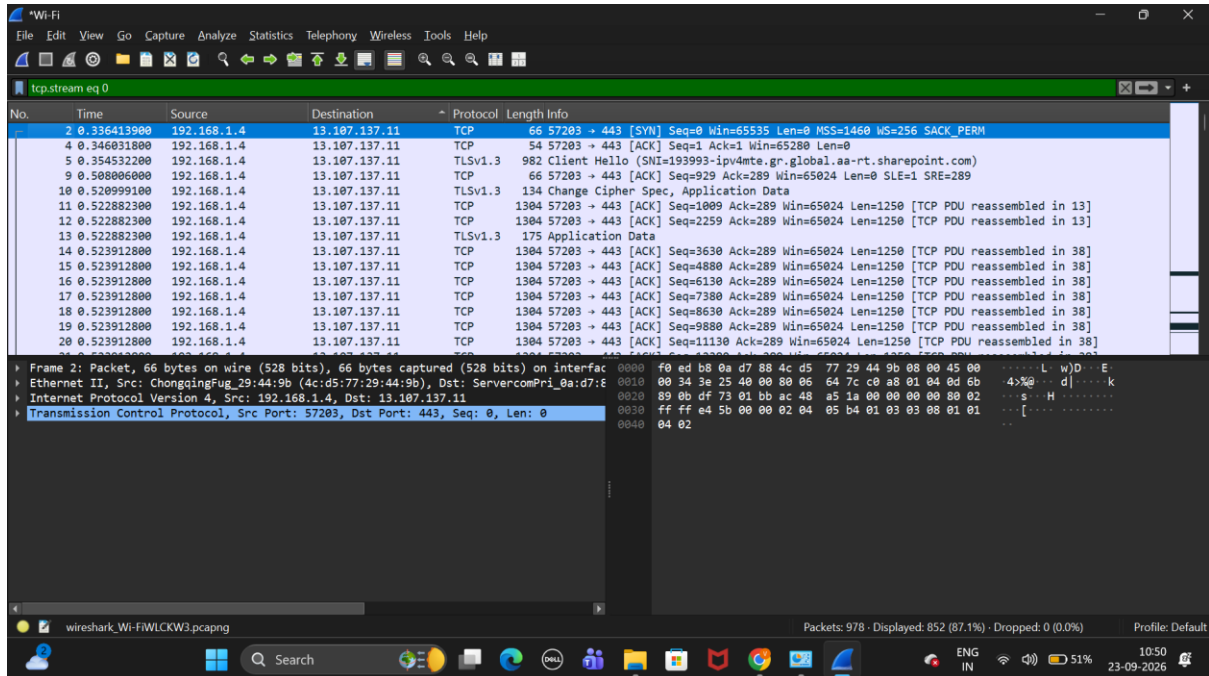


Figure 11: TCP 3-Way Handshake observed in Wireshark.

7. Summary of Findings

- DNS was used for domain name resolution.
- TCP provided reliable communication.
- TLS encrypted HTTPS traffic.
- ARP resolved IP addresses to MAC addresses.
- HTTP traffic was visible in plain text, while HTTPS traffic was encrypted.

8. Conclusion

Wireshark successfully captured and analyzed different network protocols. The experiment demonstrated how network communication occurs and highlighted the importance of encryption in securing data transmission.